

Brazilian E-Commerce Public Dataset by Olist

1 Introduction

In this assignment, we will build and compare a variety of classification models to predict whether an employee will leave the company ('left_company'), using the provided HR dataset.

2 Data

2.1 Dataset

The data files are as follows:

- train.csv (1341 rows, 35 columns): 33 feature columns + ID + target left_company
- test.csv (336 rows, 34 columns): same features + ID, no target
- sample_submission.csv (336 rows, 2 columns): template for submission (ID, left_company)

we use the train.csv to train the models.

2.2 Features

Features include numerical, ordinal, and categorical variables such as age, salary, travel frequency, job satisfaction ratings, and tenure measures.

The features are as follows:

	Column	Non-Null Count	Dtype
0	Unnamed: 0	1341 non-null	int64
1	age_years	1341 non-null	int64
2	travel_freq	1341 non-null	object

3	daily_salary	1341 non-null	int64
4	work_division	1341 non-null	object
5	commute_distance	1341 non-null	int64
6	education_level	1341 non-null	int64
7	degree_field	1341 non-null	object
8	headcount	1341 non-null	int64
9	env_satisfaction	1341 non-null	int64
10	sex	1341 non-null	object
11	hourly_wage	1341 non-null	int64
12	job_engagement	1341 non-null	int64
13	position_level	1341 non-null	int64
14	job_title	1341 non-null	object
15	job_satisfaction	1341 non-null	int64
16	marital_state	1341 non-null	object
17	monthly_earnings	1341 non-null	int64
18	monthly_salary	1341 non-null	int64
19	companies_count	1341 non-null	int64
20	is_adult	1341 non-null	object
21	overtime_status	1341 non-null	object
22	salary_increase_pct	1341 non-null	int64
23	performance_score	1341 non-null	int64
24	relation_satisfaction	1341 non-null	int64
25	std_work_hours	1341 non-null	int64
26	stock_level	1341 non-null	int64
27	career_years	1341 non-null	int64
28	trainings_last_year	1341 non-null	int64
29	work_life_score	1341 non-null	int64
30	tenure_years	1341 non-null	int64
31	years_current_role	1341 non-null	int64

32	years_post_promotion	1341 non-null	int64
33	years_with_manager	1341 non-null	int64
34	left_company	1341 non-null	int64

Then I checked the missing values which this dataset does not contain any.

Missing Values:

Unnamed: 0	0
age_years	0
travel_freq	0
daily_salary	0
work_division	0
commute_distance	0
education_level	0
degree_field	0
headcount	0
env_satisfaction	0
sex	0
hourly_wage	0
job_engagement	0
position_level	0
job_title	0
job_satisfaction	0
marital_state	0
monthly_earnings	0
monthly_salary	0
companies_count	0

is_adult	0
overtime_status	0
salary_increase_pct	0
performance_score	0
relation_satisfaction	0
std_work_hours	0
stock_level	0
career_years	0
trainings_last_year	0
work_life_score	0
tenure_years	0
years_current_role	0
years_post_promotion	0
years_with_manager	0
left_company	0

I thought finding outliers here makes no sense since the target feature only accept two values.

I also encoded the categorical features.

2.3 unique values per categorical col

travel_freq

3

work_division

3

degree_field

6

sex

2

job_title

9

marital_state

3

is_adult

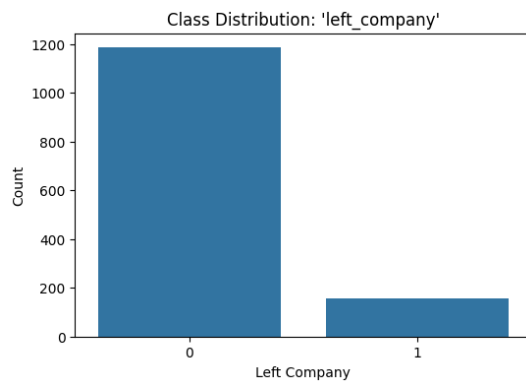
1

overtime_status

2

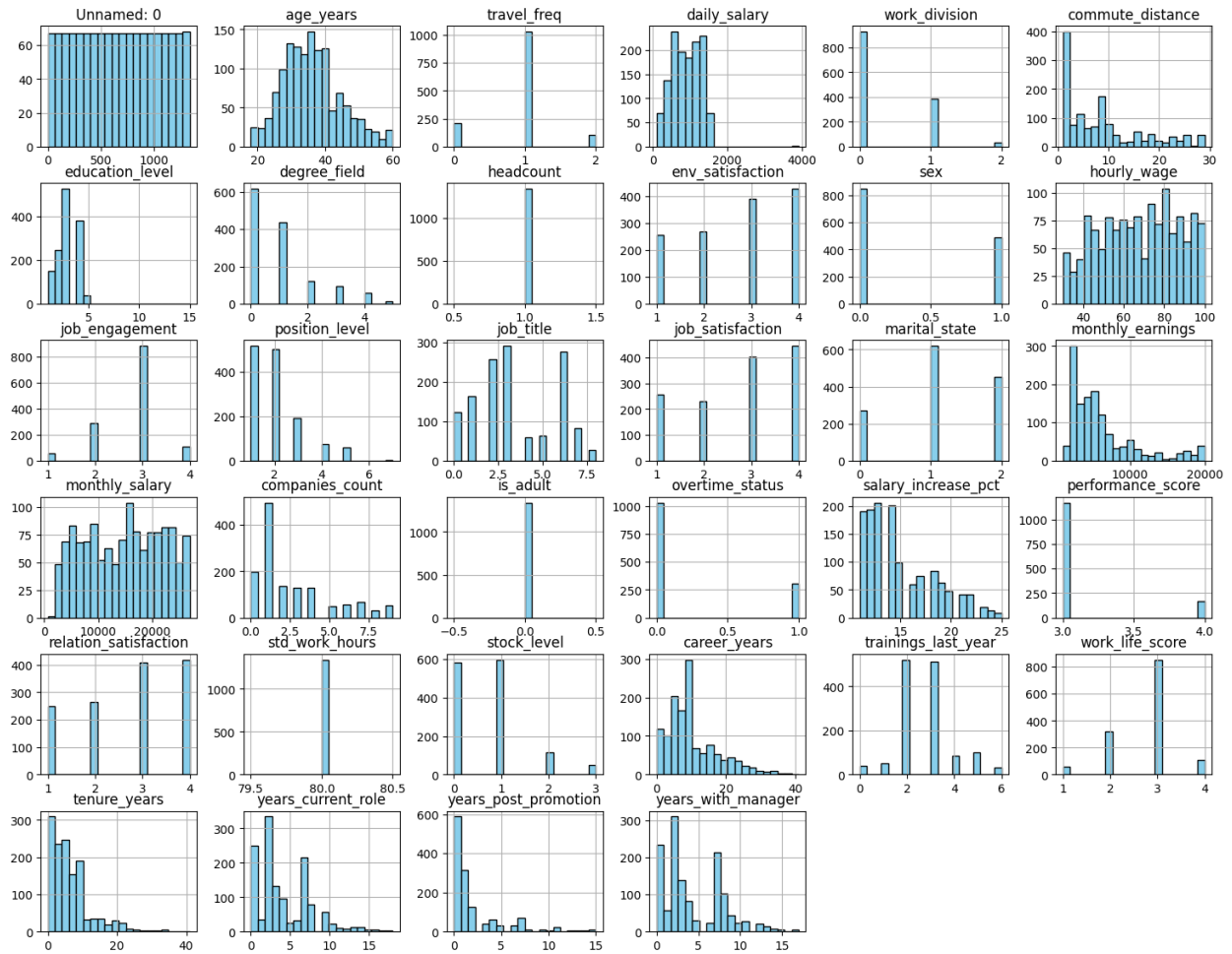
2.4 visualization

In this section we will go through some visualization related to the data that can help us to have a better sense of our data and the relationship between the features.

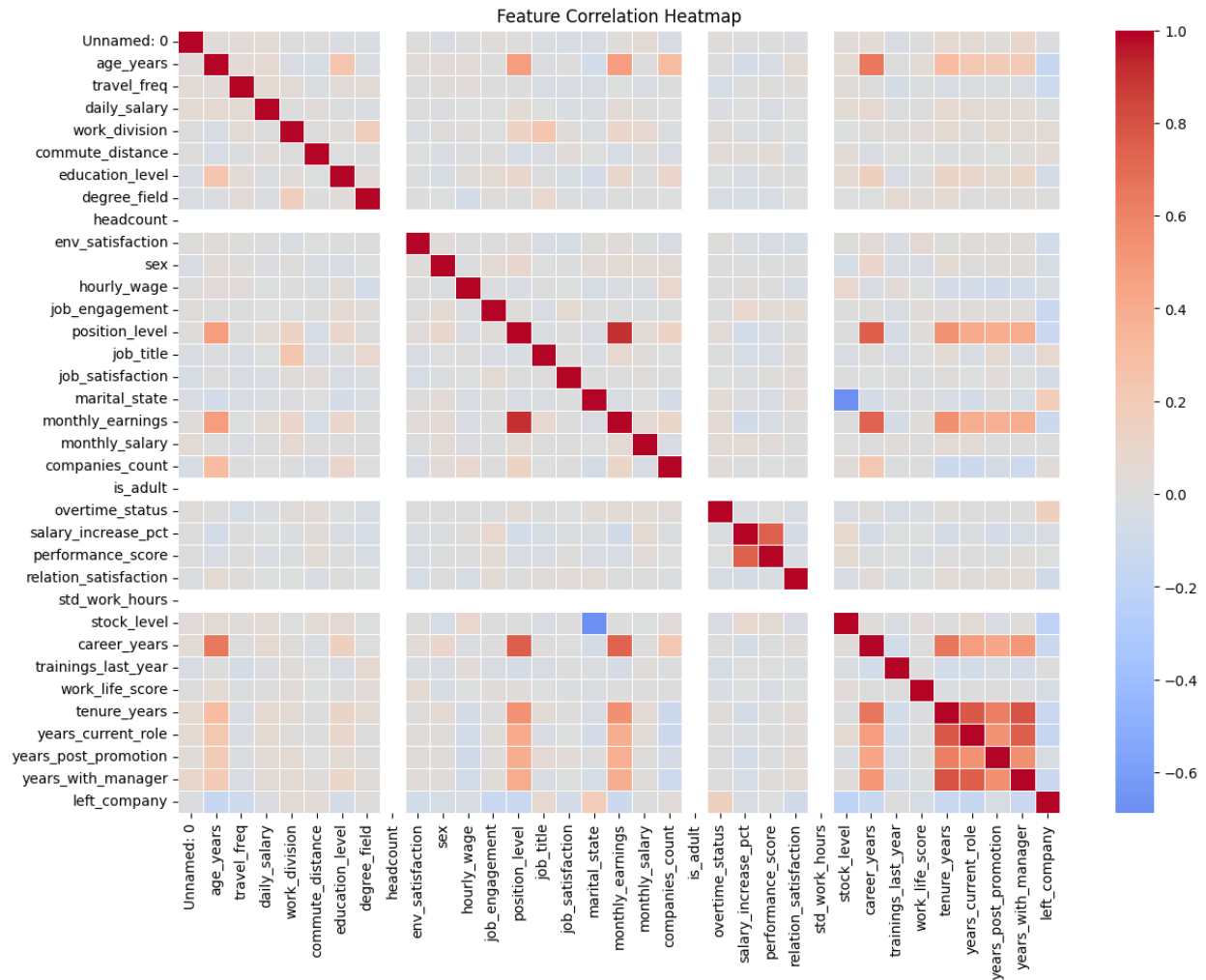


In this plot we can see the distribution of the target feature. What we can see is that the dataset is not balanced and the number of 0s are too much more than 1s.

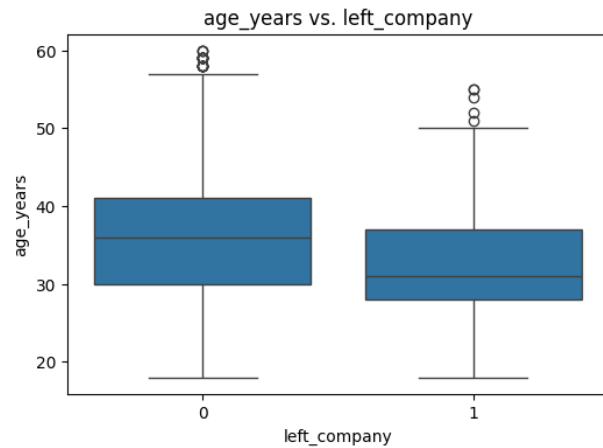
Feature Distributions



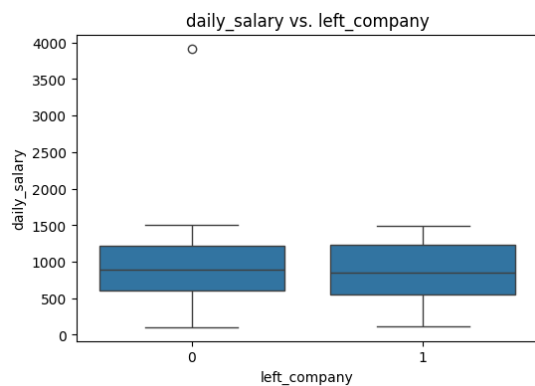
In this image we can see the plots for the distribution of different features. The distributions shows the range and frequency of these characteristics across the employee population providing valuable insights for data analysis and modeling.



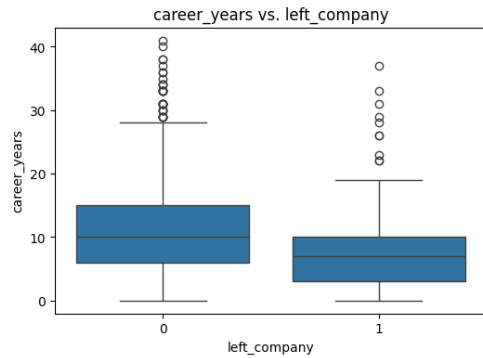
The provided image is a feature correlation heatmap that visualizes the relationships between various attributes in the employee dataset. When the color of the intersection between two feature tends to be red it shows that the positive correlation is stronger and the two features have direct relationship with each other and when the color tends to blue it shows negative strong correlation.



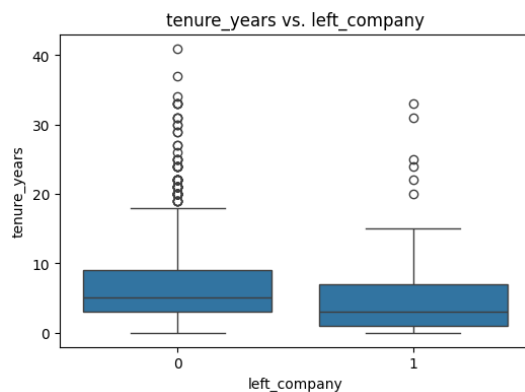
The provided image is a bar chart that shows the relationship between an employee's age in years and whether they left the company or not. As we can see the avg age for the employees who left the company is less than the ones who did not indicating that the older employees tend to be more loyal to the company and less likely to want to change their environment.



As we can see the amount of salary is not related to the employees' desire to leave or not and does not meaningfully affect it since the avg salary for both groups are almost the same.



It seems that the more the working experience the less desire to leave the company.



The employees who worked in the company tend to want to leave less.

3 Baseline Clustering

In this section a full machine learning pipeline was developed to predict whether employees would leave the company. The data was first cleaned and preprocessed by removing irrelevant columns and scaling the numerical features. Then feature correlations were explored to identify the most related variables. The dataset was split into training and validation sets using stratified sampling to preserve class distribution. Several classification models including LR, RF and SVM with balanced class weights were trained and compared using 10 fold cross validation. Their performance was measured based on accuracy training time and score variability. After that two versions of LR (with and without class weighting) were analyzed in more depth using metrics like precision, recall, F1 score and ROC AUC. Their ROC curves and confusion matrices were also visualized to better understand how they handle class imbalance.

	Name	Parameters \
1	RandomForestClassifier	{'bootstrap': True, 'ccp_alpha': 0.0, 'class_w...
0	LogisticRegression	{'C': 1.0, 'class_weight': 'balanced', 'dual':...
2	SVC	{'C': 1.0, 'break_ties': False, 'cache_size': ...

	Train Accuracy	Mean Test Accuracy	Mean Test Accuracy 3*STD	Time
1	1.0	0.881437	0.027086	0.694818
0	0.750685	0.725627	0.160732	0.83782
2	0.575277	0.570481	0.109323	0.758453

Logistic Regression (No Class Weights):

Precision: 0.6666666666666666

Recall: 0.06451612903225806

F1 Score: 0.11764705882352941

ROC AUC: 0.7258064516129032

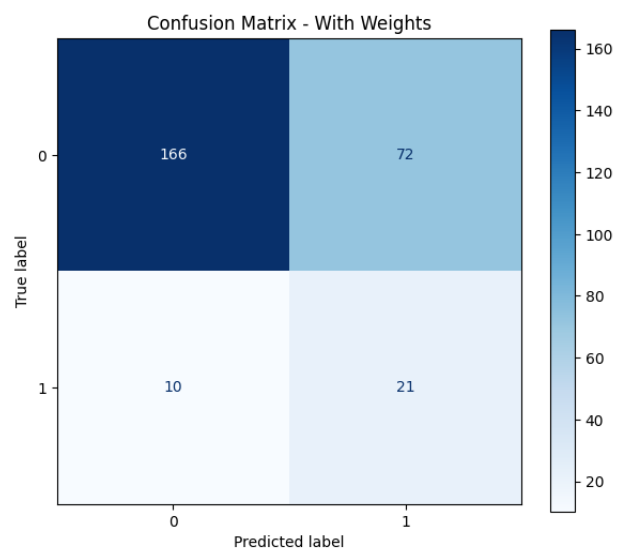
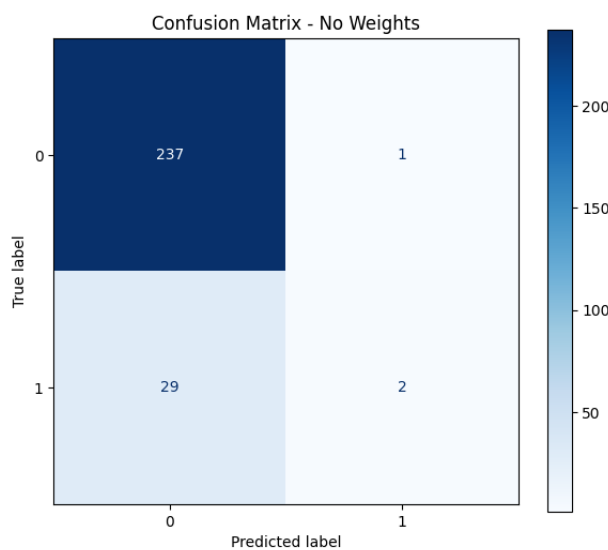
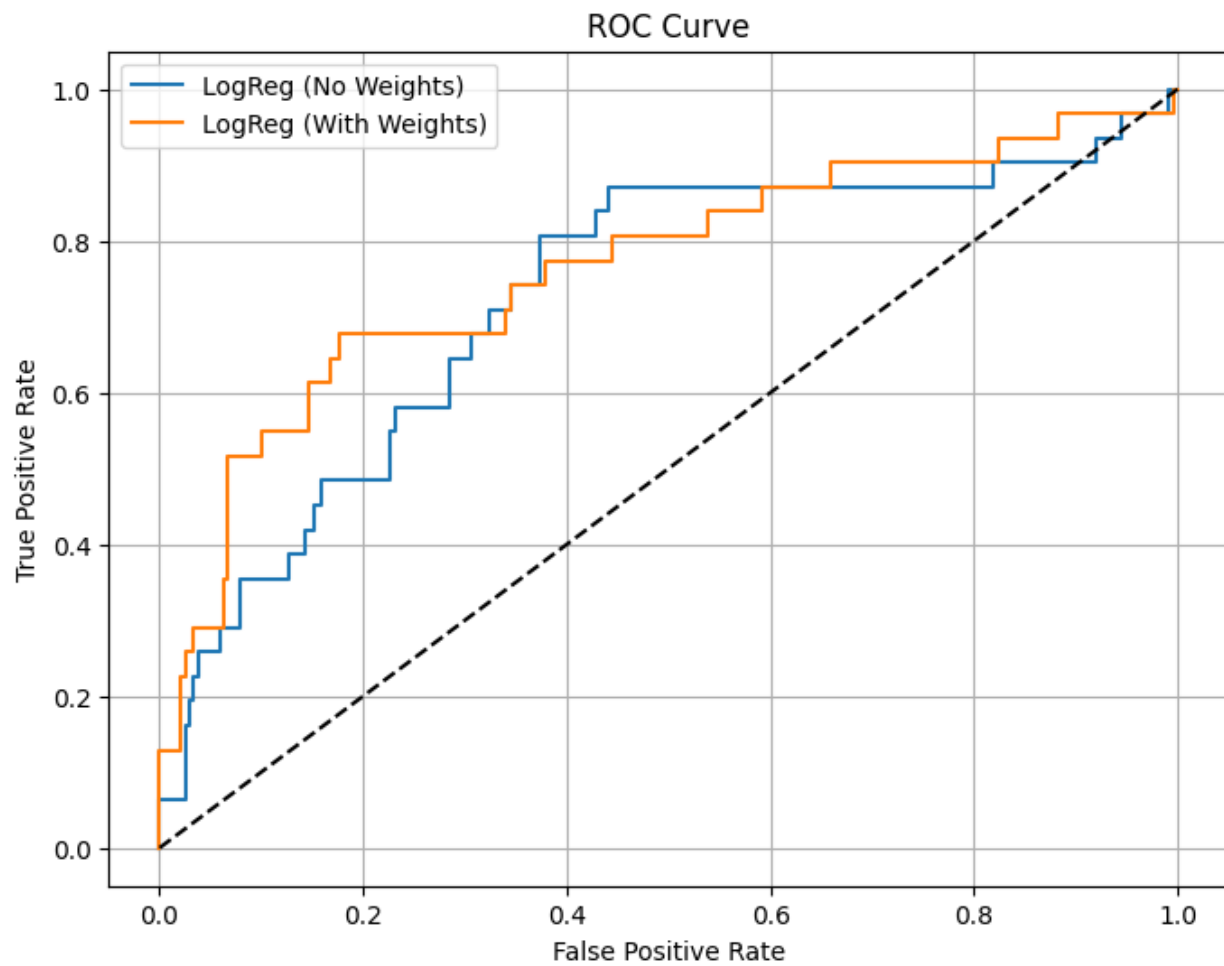
Logistic Regression (With Class Weights):

Precision: 0.22580645161290322

Recall: 0.6774193548387096

F1 Score: 0.3387096774193548

ROC AUC: 0.7637571157495255



As we can see it seems like the LR with weights seems to have a better performance as our dataset is imbalanced.

4 Advanced Models

In this part of the analysis, several classification models were trained and evaluated using pipelines that include SMOTE to handle class imbalance. Three main classifiers KNNs, Naive Bayes and SVM with an RBF kernel were first tested. Each was wrapped in a pipeline where SMOTE was applied before training. The RBF SVM was tuned using a grid search over different values of C and gamma with the best model selected based on F1 score. Additionally, a linear SVM was implemented using LinearSVC inside a CalibratedClassifierCV to obtain probability estimates. This model also underwent hyperparameter tuning with CV. All models were evaluated on a validation set using F1 score and ROC AUC and for the linear SVM a confusion matrix and ROC curve were plotted to visualize its performance.

=== SVM RBF ===

F1 Score: 0.28776978417266186

ROC AUC Score: 0.5969775006776904

=== KNN ===

F1 Score: 0.1889763779527559

ROC AUC Score: 0.5040661425860666

=== Naive Bayes ===

F1 Score: 0.34234234234234234

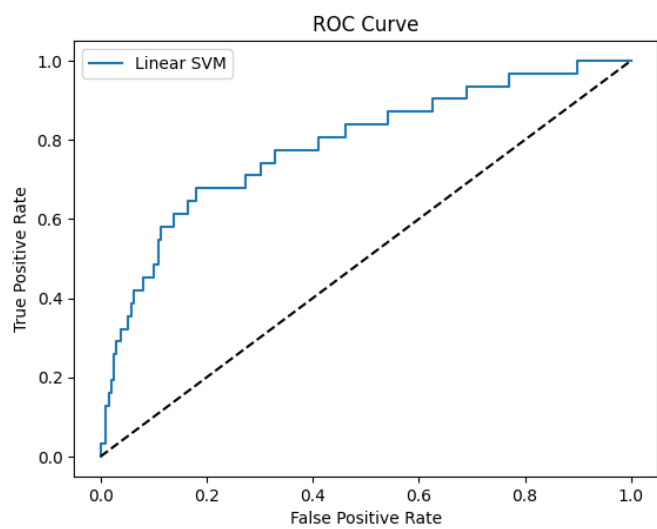
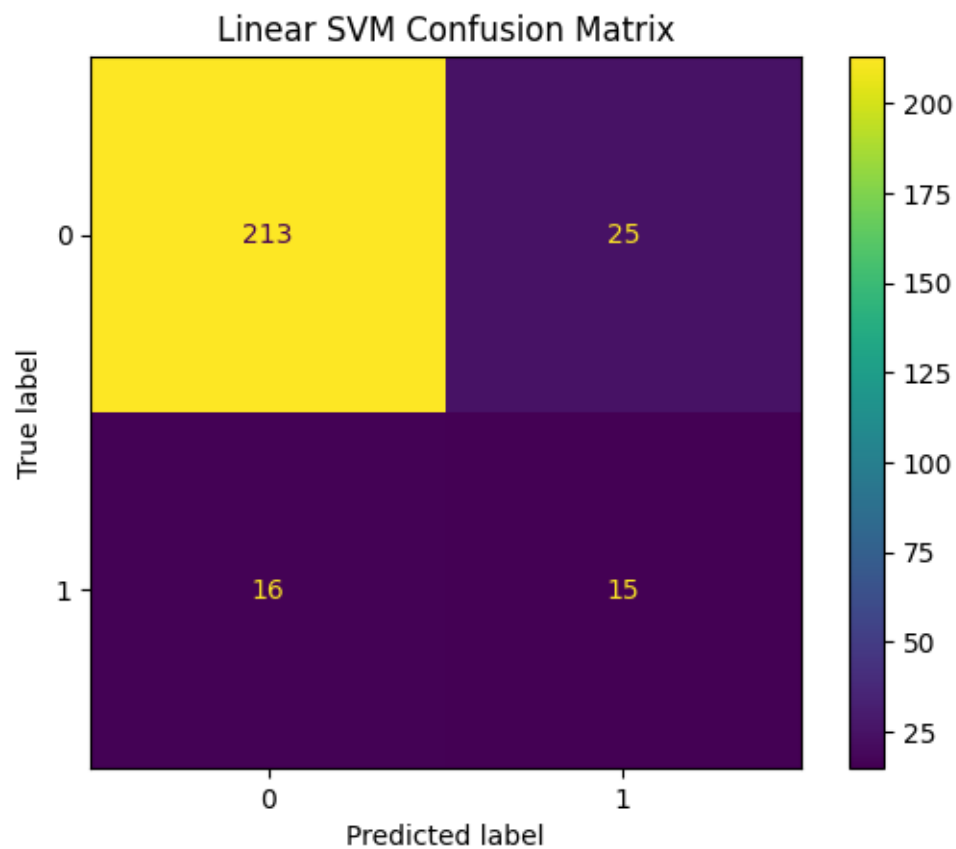
ROC AUC Score: 0.701409596096503

=== Optimized SVM Linear ===

Best Parameters: {'svc__estimator__C': 1, 'svc__estimator__class_weight': 'balanced'}

F1 Score: 0.4225352112676056

ROC AUC Score: 0.7854432095418813



5 Implementation of RF from scratch (BONUS)

In this section a custom RF classifier was implemented from scratch to better understand how ensemble learning works behind the scenes. The SimpleRandomForest class uses bootstrapped samples and random feature subsets to train multiple decision trees then aggregates their predictions via majority voting. This model was then compared to the standard RandomForestClassifier from scikit-learn which was configured with more estimators and class balancing. Besides the random forest models two other baseline classifiers kNN and Naïve Bayes were also trained and evaluated. All models were assessed on the validation set using F1 score and ROC AUC allowing for a direct performance comparison between the custom implementation and library models.

=== Simple RF (from scratch) ===

F1 Score: 0.0625

ROC AUC Score: 0.7251287611818923

=== Library RF ===

F1 Score: 0.11764705882352941

ROC AUC Score: 0.8152615885063702

=== KNN ===

F1 Score: 0.0

ROC AUC Score: 0.5372729737056112

=== Naive Bayes ===

F1 Score: 0.3953488372093023

ROC AUC Score: 0.7191650853889943

6 Handling Imbalanced Data

To address class imbalance more better, both LR and RBF SVM models were trained with the `class_weight='balanced'` option enabled. This adjusts the weights inversely proportional to class frequencies which helps the models focus more on the minority class. After fitting the models on the training data predictions and probability estimates were generated for the validation set and their performance was evaluated using F1 score and ROC AUC. Plus for the LR a custom decision threshold of 0.3 was applied instead of the default 0.5. This adjustment improves recall by lowering the threshold required to classify a sample as positive.

=== Logistic Regression (class_weight) ===

F1 Score: 0.42857142857142855

ROC AUC Score: 0.8126863648685281

=== SVM RBF (class_weight) ===

F1 Score: 0.25

ROC AUC Score: 0.6223908918406071

=== LogReg + Threshold = 0.3 ===

F1 Score: 0.3312883435582822

ROC AUC Score: 0.8126863648685281

7 model stacking (optional)

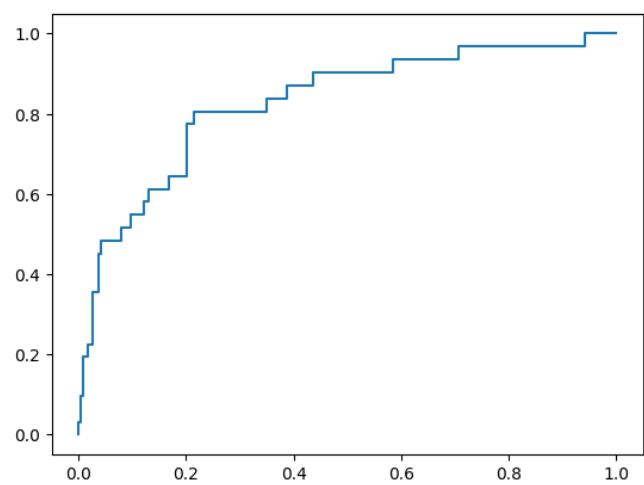
After training a bunch of individual models I wanted to try combining their strengths. First I built a stacked model using logistic regression and random forest as the base models then added another logistic regression on top to combine their outputs. I used CV to train it and checked how it did on the validation set using F1 score and ROC AUC. Then I tried a voting classifier that mixes several models: logistic regression, random forest, SVM and XGBoost. I gave a bit more weight to the models that were performing better (random forest and XGBoost). But before training it I used RFE to pick the top 30 most useful features, since not all features were equally important. Once the voting model was trained I looked at the precision-recall curve to find the best threshold for getting a good balance between precision and recall. I picked the one with the highest F1 score and used that to make final predictions.

To visualize how well the model did I plotted the confusion matrix and the precision-recall curve. After that I used the trained model to predict the test set and saved the results to a CSV file for submission.

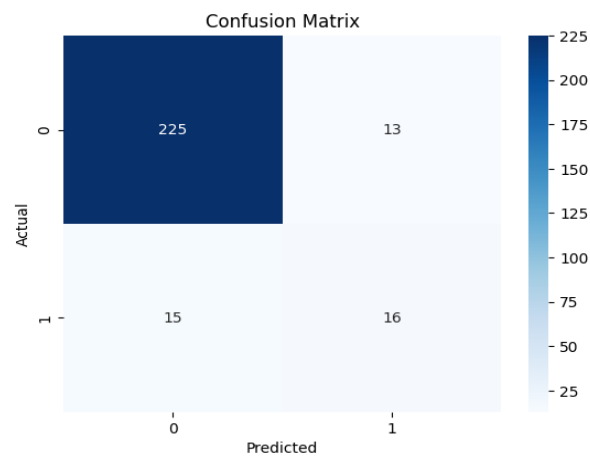
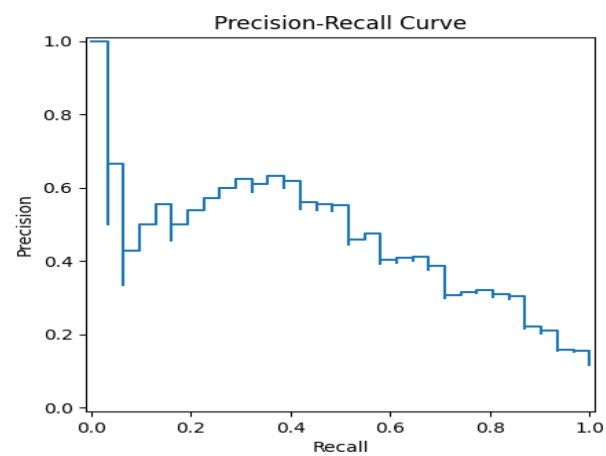
=== Stacked Model ===

F1 Score: 0.11764705882352941

ROC AUC Score: 0.8281377066955815



Best threshold by F1 score: 0.23					
	precision	recall	f1-score	support	
0	0.94	0.95	0.94	238	
1	0.55	0.52	0.53	31	
accuracy			0.90	269	
macro avg	0.74	0.73	0.74	269	
weighted avg	0.89	0.90	0.89	269	



Conclusion

In this project I experimented with various classification models from individual algorithms like logistic regression, SVM, and random forest, to ensemble techniques like stacking and voting. By using methods such as SMOTE for handling class imbalance feature selection with RFE and threshold tuning based on the precision-recall curve I was able to improve model performance significantly. The final ensemble model achieved a strong balance between precision and recall making it well suited for predicting employee attrition.

The score I received on Kaggle on public is 0.51162 and my rank at the time that I am writing this is 6.

References

<https://www.kaggle.com/competitions/ml4sbu/overview>

Maryam Soleimani 401222075